

hw1

chl

中文教材：p8, ex. 1, 6, 8, 9. p15. ex. 2, 4, 8, 12

英文教材：p19-20: ex. 1, 2, 3, 5, 6

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1.

设 A, B, C, D 是某一试验中的 4 个事件. 试用它们的运算表达如下事件:

- (1) 4 个事件中至少一个发生;
- (2) 恰好发生两个;
- (3) 4 个事件中至少发生三个;
- (4) 至多发生一个.

1. $A \cup B \cup C \cup D$

2.

$$\begin{aligned} & (A \cap B \cap \bar{C} \cap \bar{D}) \\ & \cup (A \cap \bar{B} \cap C \cap \bar{D}) \\ & \cup (A \cap \bar{B} \cap \bar{C} \cap D) \\ & \cup (\bar{A} \cap B \cap C \cap \bar{D}) \\ & \cup (\bar{A} \cap B \cap \bar{C} \cap D) \\ & \cup (\bar{A} \cap \bar{B} \cap C \cap D) \end{aligned}$$

3.

$$\begin{aligned}
 & (A \cap B \cap C \cap \overline{D}) \\
 & \cup (A \cap B \cap \overline{C} \cap D) \\
 & \cup (A \cap \overline{B} \cap C \cap D) \\
 & \cup (\overline{A} \cap B \cap C \cap D) \\
 & \cup (A \cap B \cap C \cap D)
 \end{aligned}$$

4.

$$\begin{aligned}
 & (\overline{A} \cap \overline{B} \cap \overline{C} \cap \overline{D}) \\
 & \cup (A \cap \overline{B} \cap \overline{C} \cap \overline{D}) \\
 & \cup (\overline{A} \cap B \cap \overline{C} \cap \overline{D}) \\
 & \cup (\overline{A} \cap \overline{B} \cap C \cap \overline{D}) \\
 & \cup (\overline{A} \cap \overline{B} \cap \overline{C} \cap D)
 \end{aligned}$$

6.

设事件列 $\{A_n\}$ 单调上升：即任 $n \geq 1$ 有 $A_n \subset A_{n+1}$ 。试用概率论语言证明

$$\overline{\lim}_n A_n = \bigcup_{n=1}^{\infty} A_n = \lim_n A_n.$$

对单调下降事件列，相应的结论是什么？

Proof. 显然 $\forall n, \bigcup_{k=n}^{\infty} A_k \supset \bigcup_{k=n+1}^{\infty} A_k$, 同时由于 $\{A_n\}$ 单调上升, $A_n \subset A_{n+1} \subset \bigcup_{k=n+1}^{\infty} A_k$, $\bigcup_{k=n}^{\infty} A_k \subset \bigcup_{k=n+1}^{\infty} A_k$, 故 $\bigcup_{k=n}^{\infty} A_k = \bigcup_{k=n+1}^{\infty} A_k$, 进而

$$\overline{\lim}_n A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k = \bigcap_{n=1}^{\infty} \bigcup_{k=1}^{\infty} A_k = \bigcup_{n=1}^{\infty} A_n$$

由于 $\{A_n\}$ 单调上升, $\forall n, \bigcap_{k=n}^{\infty} A_k = A_n$, 从而

$$\varliminf_n A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k = \bigcup_{n=1}^{\infty} A_n$$

对单调下降事件 $B_1 \supset B_2 \supset \cdots$, $B_1^c \subset B_2^c \subset \cdots$ 是单调上升事件, 有

$$\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} B_k^c = \overline{\varlimsup_n B_n^c} = \bigcup_{n=1}^{\infty} B_n^c = \varliminf_n B_n^c = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} B_k^c$$

等号各侧取余, 根据 *De Morgan* 对偶律, 得到

$$\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} B_k = \varlimsup_n B_n = \bigcap_{n=1}^{\infty} B_n = \overline{\varliminf_n B_n} = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} B_k$$

□

8.

画出表示下列各事件间关系的 Veen 图

(1) 将一枚硬币抛三次,

$A = \{\text{第一次为正面}\},$

$B = \{\text{3次出现同一面}\},$

$C = \{\text{有正面}\};$

(2) 将一枚骰子掷两次,

$A = \{\text{点数相同}\},$

$B = \{\text{其中一枚点数是另一枚的 2 倍}\},$

$C = \{\text{点数之和为 6}\};$

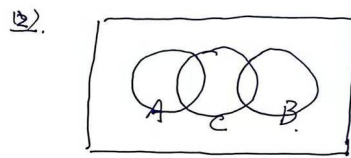
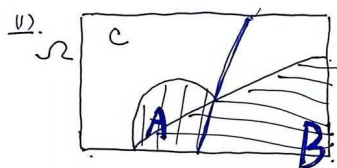
(3) 将一红球与一黑球任放入编号为 1,2,3 的三个盒中,

$A = \{\text{1号盒不空}\},$

$B = \{\text{1号盒与 2 号盒各一球}\},$

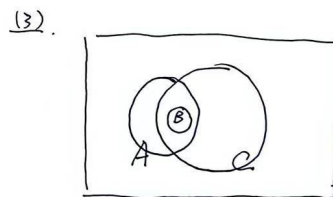
$C = \{\text{每盒至多一球}\}.$

8. (1) $A: H _ _$, $AB: HHH$, $A \cup B \cup C = \Omega$.
 $B: HHH / TTT$, $AC = A$.
 $C: (H)$, $BC = HHH = ABC$.



(2) $A: 11, 22, \dots, 66$, $AB = \emptyset = ABC$, $A \cup B \cup C \neq \Omega$.
 $B: 12, 21, 24, 42, 36, 63$, $AC = 33$.
 $C: 15, 24, 33, 42, 51$, $BC = 24, 42$.

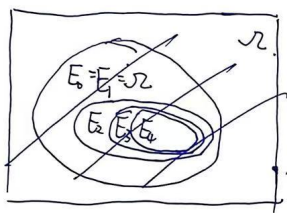
(3) $AB = B$, $1R2B, 1B2R$,
 $AC = 1R2B, 1B2R, 1R3B, 1B3R$,
 $BC = 1B$, $ABC = B$, $A \cup B \cup C \neq \Omega$.



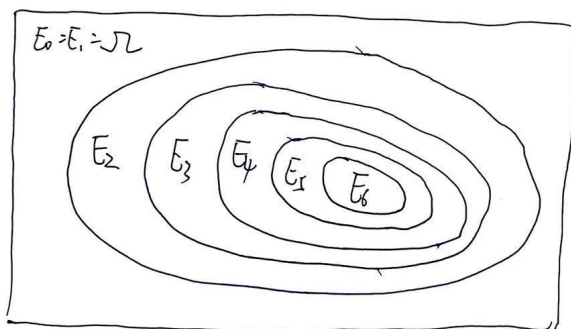
9.

从 0 至 9 这 10 个数字中有重复地取 5 个. 以 E_i 表示有某数字被取出 i 次 (例如取出 52353, 则 E_0, E_1, E_2, E_3 三事件均发生). 试画出表示 $E_0, E_1, E_2, E_3, E_4, E_5, E_6$ 之间关系的 Venn 图.

9. 若某数字出现 i 次, 那它一定出现了 $0 \sim i-1$ 次. 故 $E_0 \supset E_1 \supset E_2 \supset \dots$



$$E_0 = U = E_1 \supset E_2 \supset E_3 \supset \dots$$



2.

对于自然数 $r \geq n$, 设计适当的试验, 求不定方程 $x_1 + x_2 + \dots + x_n = r$ 的:

(1) 非负整数解的个数; (2) 正整数解的个数.

Proof. 将 r 看作 r 个 1, 在相邻的 1 之间插入隔板, 相邻隔板间 1 的个数对应 x_i 的值, 这对应正整数解的个数, 为 $\binom{r-1}{n-1}$. 对于非负整数解, 我另外拿 n 个 1 过来添给每个 x_i , 即令 $b_i = x_i + 1$, 于是等价于 $\sum_{i=1}^n b_i = n + r$ 的正整数解个数, 为 $\binom{n+r-1}{n-1}$. $\square$

4.

从 n 对不同的鞋子中任取出 $2m$ 只 ($2m < n$)，求其中恰有 k 双配对的鞋子的概率.

Proof.

$$\Omega = \{n \text{ 对不同的鞋子中任取出 } 2m \text{ 只}\} \quad |\Omega| = \binom{2n}{2m}$$

$$E = \{\text{恰有 } k \text{ 双配对}\} \quad |E| = \binom{n}{k} \binom{n-k}{2m-2k} \cdot 2^{2m-2k}$$

先找出哪 k 对配对 $\binom{n}{k}$ ，然后还要取 $2m-2k$ 只鞋子，每只对应原来某一对中的某一只，于是从剩余 $n-k$ 对中找到 $2m-2k$ 需要贡献一只鞋的对 $\binom{n-k}{2m-2k}$ ，每队可以贡献其中一只鞋，有 2^{2m-2k} 种

$$\mathfrak{P} = \frac{2^{2m-2k} \binom{n}{k} \binom{n-k}{2m-2k}}{\binom{2n}{2m}}$$

□

8.

将 r 个相同的球任放入编号为 1 至 n 的 n 个盒 ($r \leq n$)，每球入各盒均等可能. 试求如下事件的概率:

- (1) 指定 r 个盒恰各 1 球;
- (2) 每盒至多 1 球;
- (3) 指定盒恰有 m 个球;
- (4) 恰有 m 个空盒.

Proof.

$$\Omega = \{r \text{ 个球放 } n \text{ 个盒}\} \quad |\Omega| = n^r$$

1.

$$|E| = r!, \quad \mathfrak{P} = \frac{r!}{n^r}$$

2. 选取 r 个盒子，这 r 个盒子恰有 1 球

$$|E| = \binom{n}{r} r!, \quad \mathfrak{P} = \frac{\binom{n}{r} r!}{n^r}$$

3. $r - m$ 个球放剩余 $n - 1$ 个盒

$$|E| = \frac{r}{m} (n - 1)^{r-m}, \quad \mathfrak{P} = \frac{\frac{r}{m} (n - 1)^{r-m}}{n^r}$$

4. 挑 m 个空盒, r 个球放 $n - m$ 个盒, 每盒至少一个, 对这些盒子重新编号, 令

$$A_i = \{\text{盒 } i \text{ 是空的}\} \quad \text{所求} = \bigcap_{i=1}^{n-m} A_i^c = \bigcup_{i=1}^{n-m} A_i$$

根据容斥原理,

$$\text{所求} = \sum_{j=0}^{n-m} (-1)^j \binom{n-m}{j} (n-m-j)^r$$

从而

$$\mathfrak{P} = \frac{\binom{n}{m} \sum_{j=0}^{n-m} (-1)^j \binom{n-m}{j} (n-m-j)^r}{n^r}$$

□

12.

袋中装有编号为 1 至 N 的 N 个球, 现有放回地任取 n 个, 依次记下其号码, 求:

(1) n 个号码按严格上升排列的概率;

(2) n 个号码非降排列的概率.

Proof. 所有可能的情形数为

$$|\Omega| = N^n$$

1. 任意选定的 n 个球，要求严格上升必须标号互不相同，且 n 个标号按升序只有一种排列故

$$|E| = \binom{N}{n} \quad \mathfrak{P} = \frac{\binom{N}{n}}{N^n}$$

2. 对于非降序列， n 个球的编号可表示为

$$1 \leq a_{1,1} = \cdots = a_{1,r_1} < a_{2,1} = \cdots = a_{2,r_2} < \cdots < a_{t,1} = \cdots = a_{t,r_t} \leq N$$

其中 $1 \leq t \leq n \wedge N$, 对每个 t , 从 $1 \sim N$ 中选出 t 个数，此时可能的情形为

$$r_1 + \cdots + r_t = n$$

的正整数解个数，因此

$$|E| = \sum_{t=1}^{n \wedge N} \binom{N}{t} \binom{n-1}{t-1} = \sum_{t=1}^{n \wedge N} \binom{N}{t} \binom{n-1}{n-t}$$

考虑如下情形：从 N 元集合 A 和 $n-1$ 元集合 B 中分别取总数为 n 的代表元，并要求取出的代表元中必须同时含有 A 和 B 的元素. 若 $N < n$, 则情况数即为上式右侧，此时 A 、 B 元素数均小于所取代表元数，因此所求情况数亦可表示为 $\binom{N+n-1}{n}$. 若 $N \geq n$, 则情况数为 $\sum_{t=1}^{n-1} \binom{N}{t} \binom{n-1}{n-t} = \sum_{t=1}^{n \wedge N} \binom{N}{t} \binom{n-1}{n-t} - \binom{N}{n}$, 同时所求情况总数亦可表示为 $\binom{N+n-1}{n} - \binom{N}{n}$. 这两种情形都导出 $|E| = \binom{N+n-1}{n}$, 从而

$$\mathfrak{P} = \frac{\binom{N+n-1}{n}}{N^n}.$$

事实上，可以考虑每个编号 i 出现的次数 c_i ，一旦这个确定了， n 个球编号的情况也就确定了（按升序排列即可），故只需考虑

$$c_1 + \cdots + c_N = n$$

的非负整数解数，即 $\binom{N+n-1}{N-1} = \binom{N+n-1}{n}$, 从而

$$\mathfrak{P} = \frac{\binom{N+n-1}{n}}{N^n}.$$

□

***1.**

$(\bigcup_j A_j) \setminus (\bigcup_j B_j) \subset \bigcup_j (A_j \setminus B_j) \cdot (\bigcap_j A_j) \setminus (\bigcap_j B_j) \subset \bigcup_j (A_j \setminus B_j)$. When is there equality?

Proof.

$$\begin{aligned}
\forall x \in (\bigcup_j A_j) \setminus (\bigcup_j B_j) &\implies \exists i, x \in A_i \quad \forall j, x \notin B_j \implies x \in A_i \setminus B_i \\
&\implies x \in \bigcup_j (A_j \setminus B_j) \implies (\bigcup_j A_j) \setminus (\bigcup_j B_j) \subset \bigcup_j (A_j \setminus B_j) \\
\forall x \in (\bigcap_j A_j) \setminus (\bigcap_j B_j) &\implies \forall i, x \in A_i \quad \exists j, x \in B_j \implies x \in A_j \setminus B_j \\
&\implies x \in \bigcup_j (A_j \setminus B_j) \implies (\bigcap_j A_j) \setminus (\bigcap_j B_j) \subset \bigcup_j (A_j \setminus B_j)
\end{aligned}$$

If there is equality,

1.

$$\begin{aligned}
(\bigcup_j A_j) \setminus (\bigcup_j B_j) \supset \bigcup_j (A_j \setminus B_j) &\implies \forall x \in \bigcup_j (A_j \setminus B_j), \exists j, x \in A_j \setminus B_j \\
&\implies (\surd)x \in \bigcup_j A_j, \quad (?)x \notin \bigcup_j B_j
\end{aligned}$$

This means as long as $x \in$ some $A_j \setminus B_j$, then $x \notin \bigcup_j B_j$, which means

$$\left(\bigcup_j A_j \setminus B_j \right) \cap \left(\bigcup_j B_j \right) = \emptyset$$

2.

$$\begin{aligned} (\bigcap_j A_j) \setminus (\bigcap_j B_j) &\supset \bigcup_j (A_j \setminus B_j) \implies \forall x \in \bigcup_j (A_j \setminus B_j), \exists j, x \in A_j \setminus B_j \\ \implies (?) x &\in \bigcap_j A_j \quad (\checkmark) x \notin \bigcap_j B_j \end{aligned}$$

This means as long as $x \in$ some $A_j \setminus B_j$, then $x \in \bigcap_j A_j$, which means

$$\bigcup_j A_j \setminus B_j \subset \bigcap_j A_j$$

Then $\forall j$,

$$A_j \setminus B_j \subset \bigcap_j A_j \iff A_j \setminus \left(\bigcap_j A_j \right) \subset B_j$$

□

***2.**

The best way to define the symmetric difference is through indicators of sets as follows:

$$1_{A \triangle B} = 1_A + 1_B \pmod{2}$$

where we have arithmetical addition modulo 2 on the right side. All properties of $\triangle$ follow easily from this definition, some of which are rather tedious to verify otherwise. As examples:

$$(A \triangle B) \triangle C = A \triangle (B \triangle C),$$

$$(A \triangle B) \triangle (B \triangle C) = A \triangle C,$$

$$(A\Delta B)\Delta(C\Delta D) = (A\Delta C)\Delta(B\Delta D),$$

$$A\Delta B = C \Leftrightarrow A = B\Delta C,$$

$$A\Delta B = C\Delta D \Leftrightarrow A\Delta C = B\Delta D.$$

Proof. Under mod 2:

1.

$$1_{(A\Delta B)\Delta C} = 1_{A\Delta B} + 1_C = 1_A + 1_B + 1_C$$

$$1_{A\Delta(B\Delta C)} = 1_A + 1_{B\Delta C} = 1_A + 1_B + 1_C$$

2.

$$1_{(A\Delta B)\Delta(B\Delta C)} = 1_{(A\Delta B)} + 1_{B\Delta C} = 1_A + 2 \cdot 1_B + 1_C = 1_A + 1_C = 1_{A\Delta C}$$

3.

$$1_{(A\Delta B)\Delta(C\Delta D)} = 1_{A\Delta B} + 1_{C\Delta D} = 1_A + 1_B + 1_C + 1_D$$

$$1_{(A\Delta C)\Delta(B\Delta D)} = 1_{A\Delta C} + 1_{B\Delta D} = 1_A + 1_C + 1_B + 1_D$$

4.

$$A\Delta B = C \iff 1_{A\Delta B} = 1_C \iff 1_A + 1_B = 1_C$$

$$\iff 1_A = 1_B + 1_C \iff 1_A = 1_{B\Delta C} \iff A = B\Delta C$$

5.

$$A\Delta B = C\Delta D \iff 1_{A\Delta B} = 1_{C\Delta D} \iff 1_A + 1_B = 1_C + 1_D$$

$$\iff 1_A + 1_C = 1_B + 1_D \iff 1_{A\Delta C} = 1_{B\Delta D} \iff A\Delta C = B\Delta D$$

□

3.

If Ω has exactly n points, then $\mathcal{F}$ has 2^n members. The B.F. generated by n given sets “without relations among them” has 2^{2^n} members.

Proof. Any subset of Ω is the union of finite elements from Ω , therefore $\mathfrak{F} = 2^\Omega$, $|\mathfrak{F}| = |2^\Omega| = 2^{|\Omega|} = 2^n$.

Let $A_1, \dots, A_n \subset \Omega$ be n sets with “no relations among them”, define the 2^n sets

$$E_\varepsilon := \bigcap_{i=1}^n B_i(\varepsilon_i), \quad \varepsilon = (\varepsilon_1, \dots, \varepsilon_n) \in \{0, 1\}^n \quad E_\varepsilon \neq \emptyset$$

among which $B_i(0) = A_i^c$ $B_i(1) = A_i$.

$$\forall \varepsilon \quad E_\varepsilon \cap E_\eta = \emptyset \quad (\varepsilon \neq \eta), \quad \bigcup_{\varepsilon \in \{0,1\}^n} E_\varepsilon = \Omega,$$

Let $\mathcal{B}$ be the **B.F.** generated by $A_1, \dots, A_n$. Every set in $\mathcal{B}$ is obtained from the A_i by c or countably many $\cup$, then

$$\forall B \in \mathcal{B}, \exists S \subseteq \{0, 1\}^n \text{ such that } B = \bigcup_{\varepsilon \in S} E_\varepsilon.$$

Therefore the map

$$g : 2^{\{0,1\}^n} \rightarrow \mathcal{B} \quad S \mapsto \bigcup_{\varepsilon \in S} E_\varepsilon$$

is a bijection. Hence

$$|\mathcal{B}| = |2^{\{0,1\}^n}| = 2^{|\{0,1\}^n|} = 2^{2^n}.$$

□

5.

The intersection of any collection of B.F.'s $\{\mathcal{F}_\alpha, \alpha \in A\}$ is the maximal B.F. contained in all of them; it is indifferently denoted by $\bigcap_{\alpha \in A} \mathcal{F}_\alpha$ or $\bigwedge_{\alpha \in A} \mathcal{F}_\alpha$.

Proof. Since it is the maximal possible collection contained in all of the $\mathfrak{F}_\alpha$, we only need to prove that $\bigcap_{\alpha \in A} \mathfrak{F}_\alpha$ is a B.F. $\forall E \in \bigcap_{\alpha \in A} \mathfrak{F}_\alpha$, then $\forall \alpha \in A$, $E \in \mathfrak{F}_\alpha$, $E^c \in \mathfrak{F}_\alpha$, $E^c \in \bigcap_{\alpha \in A} \mathfrak{F}_\alpha$. $\forall \{A_n\}_{n \geq 1} \in \bigcap_{\alpha \in A} \mathfrak{F}_\alpha$, $\forall \alpha \in A$, $\forall n$, $A_n \in \mathfrak{F}_\alpha$, $\bigcup_{n=1}^\infty A_n \in \mathfrak{F}_\alpha \implies \bigcup_{n=1}^\infty A_n \in \bigcap_{\alpha \in A} \mathfrak{F}_\alpha$. $\square$

*6.

The union of a countable collection of B.F.'s $\{\mathcal{F}_j\}$ such that $\mathcal{F}_j \subset \mathcal{F}_{j+1}$ need not be a B.F., but there is a minimal B.F. containing all of them, denoted by $\bigvee_j \mathcal{F}_j$. In general $\bigvee_{\alpha \in A} \mathcal{F}_\alpha$ denotes the minimal B.F. containing all $\mathcal{F}_\alpha, \alpha \in A$. [HINT: Ω = the set of positive integers; $\mathcal{F}_j$ = the B.F. generated by those up to j .]

Proof. Consider $\Omega = \bigcup_{n=1}^\infty \{n\}$, we define

$$\mathfrak{F}_j = \sigma \left(\bigcup_{n=1}^j \{n\} \right) \triangleq \{X, X \cup \{j+1, j+2, \dots\} : X \in 2^{\{1, 2, \dots, j\}}\} \subset 2^\Omega$$

We write $T_j \triangleq \{j+1, j+2, \dots\}$, $K_j \triangleq \{1, 2, \dots, j\}$,

1. $\forall 1 \leq k \leq j$, $\{k\} \in \mathfrak{F}_j$.
2. $\Omega = K_j \cup T_j \in \mathfrak{F}_j$.
3. $\forall A \in \mathfrak{F}_j$. If $A = B \cup T_j$, $B \subset K_j$, $A^c = K_j \setminus B \in \mathfrak{F}_j$; if $A \subset K_j$, $A^c = (K_j \setminus A) \cup T_j \in \mathfrak{F}_j$.

4. $\{A_n = B_n \cup T_j \text{ or } B_n\}_{n=1}^\infty \in \mathfrak{F}_i, B_n \in K_j,$

$$\bigcup_{n=1}^\infty A_n = \left(\bigcup_{n=1}^\infty B_n \right) \cup T_j \text{ or } \left(\bigcup_{n=1}^\infty B_n \right) \in \mathfrak{F}_j$$

5. $\forall A \in \mathfrak{F}_j$. If $A = B \cup T_j = (B \cup \{j+1\}) \cup T_{j+1}, B \subset K_j, B \cup \{j+1\} \subset K_{j+1}, A \in \mathfrak{F}_{j+1};$
if $A \subset K_j \subset K_{j+1}, A \in \mathfrak{F}_{j+1}.$

Therefore, $\{\mathfrak{F}_i\}$ is a countable collection of B.F.'s such that $\mathfrak{F}_j \subset \mathfrak{F}_{j+1}$. However, $\bigcup_{n=1}^\infty \mathfrak{F}_j = \{A \in 2^\Omega : |A| < \infty \text{ or } |A^c| < \infty\}$ does not contain $\{2n : n \in \mathbb{N}\}$. However, $\forall n \in \mathbb{N}, 2n \in \Omega$, any B.F. containing Ω must contain $\bigcup_{n=1}^\infty \{2n\} = \{2n : n \in \mathbb{N}\}.$

Considering

$$\mathcal{G} = \{A : A \text{ is B.F. and } \mathfrak{F}_j \in A, \forall j\}$$

We know that $2^\Omega \in \mathcal{G}$, so $\mathcal{G}$ is non-empty, then we can say

$$\bigcap_{j=1}^\infty \mathfrak{F}_j = \bigcap_{A \in \mathcal{G}} A$$

□